FINAL PROJECT REVIEW & DEFENSE DOSSIER • ACADEMIC YEAR 2026

AegisOne AI: An AI Copilot for Secure Web Application Development

Using Explainable Risk Analysis, Security Knowledge Graph, and Sandbox Validation

★ THE 30-SECOND ELEVATOR PITCH (WHAT TO SAY IN FRONT OF REVIEWERS)

"Traditional cybersecurity scanners drown developers in hundreds of raw warnings without explaining real business risk. When developers use tools like ChatGPT to fix them, the AI often generates code that *looks* correct but introduces hidden security holes. **AegisOne AI** is a closed-loop security copilot: it discovers the bug, explains *why* it matters using a Knowledge Graph, generates a patch, and **actually runs the patch in an isolated Docker sandbox** to prove it is safe before it ever touches production. If an AI patch is flawed, AegisOne automatically catches and rolls it back."

1. Core Problems in Application Security & How AegisOne Solves Them

Problem 1: Raw CVSS Severity Trap

Traditional tools (SonarQube, Burp) rank bugs solely by theoretical CVSS score. An SQL injection in an internal demo script gets flagged with the same panic as one in an internet-facing user database, leading to developer alert fatigue.

✓ Solution: Security Knowledge Graph & Contextual Risk Engine

AegisOne correlates static code taint sinks with runtime network exposure and data sensitivity to calculate true contextual business risk.

Problem 2: Unchecked AI Code Regressions

Generative AI (ChatGPT/Copilot) produces "plausible" fixes that hallucinate syntax errors, break API contracts, or inadvertently disable authentication routines.

✓ Solution: 4-Stage Isolated Docker Sandbox Validation

Patches run in disposable containers tested across Syntax, Regression, AST Taint, and Security Rescan. Flawed fixes are automatically rejected.

Problem 3: Rogue Out-of-Scope Scanners

Automated AI-driven scanners can inadvertently hit unauthorized third-party domains, creating serious legal liabilities and compliance violations.

✓ Solution: Hardened Scope Guard Whitelist Boundary

Architectural interceptor strictly enforces zero external network calls; only pre-approved localhost/lab endpoints are permitted.

Problem 4: Unverifiable Compliance Logs

Audit trails in traditional DevOps are easily modified or forged, making zero-trust compliance audits (OWASP/PCI-DSS) unreliable.

✓ Solution: Cryptographic SHA-256 Hash Chain

Every scan, AI suggestion, and sandbox validation run is linked in an immutable SHA-256 block chain with Ed25519 digital signatures.

2. How the Cybersecurity Mechanism Works (Step-by-Step)

AegisOne AI unifies **Static Application Security Testing (SAST)**, **Dynamic Application Security Testing (DAST)**, and **Graph Theory**:

Mechanism	How It Works Technically	Why It Is Defensible in Review
1. AST Taint Tracing (SAST)	Parses code into an Abstract Syntax Tree (AST). Traces user-controlled input (Source: `request.args.get('id')`) to execution without sanitization (Sink: `cursor.execute(...)`).	Proves deterministic evidence. We pinpoint the exact file (`app.py`) and line number (Line 42) where unsanitized data reaches SQL engine.
2. Active Fuzzing (DAST)	Dispatches non-destructive HTTP payloads (e.g. `' OR '1'='1`) against the running local lab (:8001) to verify live response behavior.	Eliminates false alarms by proving the vulnerability is actively exploitable over HTTP, not just dormant code.
3. Knowledge Graph Linkage	Constructs directed graph linking Endpoint → Taint Sink → Database Entity → CWE Definition → OWASP Category → CISA KEV Exploits.	Calculates contextual risk using Dijkstra reachability and centrality metrics rather than static CVSS numbers.
4. Closed-Loop Sandbox Validation	Executes candidate patch in a disposable container through 4 rigorous gates: `Syntax Check`, `Unit Tests`, `AST Taint Check`, and `Rescan`.	**The Showstopper**: Guarantees zero code regressions. If an AI patch causes a syntax error or contract violation, it is instantly rolled back.
5. Immutable Audit Chain	Each event computes `hash = SHA256(prev_hash + event_data + timestamp)` signed with Ed25519 key.	Provides non-repudiation. Reviewers can verify the mathematical validity of the audit chain with one click.

3. The Ten-Stage Security Decision Pipeline

When you click **"Trigger Scan"**, AegisOne AI executes a strict 10-stage sequential defense pipeline:

```
[01. DETECT] → AST pattern search & HTTP route fuzzing identify potential taint sinks.
[02. CONTEXTUALIZE] → Resolves asset exposure (public vs internal) and database sensitivity.
[03. PRIORITIZE] → Knowledge Graph scores risk (0-100); critical assets prioritized.
[04. SYNTHESIZE] → AI Security Analyst synthesizes root-cause and exploitability.
[05. PATCH] → Generates parameterized query replacement for dynamic string concatenation.
[06. VALIDATE] → Disposable Docker container environment is provisioned.
[07. ISOLATE] → Candidate code patch is hot-loaded inside isolated container boundary.
[08. REGRESSION] → Executes regression test suite to verify existing features still work.
[09. APPROVE] → If all 4 validation gates PASS, patch is stamped "Production-Eligible".
[10. SECURE] → Cryptographic SHA-256 block added to immutable audit trail.
```

4. All 20 Pages: Visual Layout & Functional Explanation

AegisOne AI contains 20 canonical pages built with the **Aegis Sentinel Dark** design system. Below is what each page does and what to explain during the review:

#	Page & Route	Visual Layout & Appearance	Functionalities & Interactive Buttons
1	**Cinematic Landing Page** `/`	Cinematic hero with high-contrast glowing cyber sphere, tactical red/cyan HUD badges, 4 security pillars, side-by-side code diff, and live pipeline infographic.	• **"Open Command Center"** routes directly to `/console`. • Interactive tabs (`Overview`, `Findings`, `Remediation`, `Validation`) switch live simulator views.
2	**Command Center** `/console`	Tactical operations dashboard with 4 metric cards (Endpoints, High Risks, Verified Patches, Compliance Rate) and real-time vulnerability feed.	• **"TRIGGER SCAN"** launches hybrid assessment. • **"EXPORT TELEMETRY"** dumps JSON system metrics. • Filter links to jump directly to critical findings.
3	**Application Inventory** `/console/inventory`	Attack surface inventory matrix displaying service names, endpoints, criticality tiers, network exposures, and data sensitivities.	• **"Add Application"** modal for new lab targets. • Exposure filters (Public vs Localhost). • Criticality badges reflecting business impact.
4	**Scan Configuration** `/console/scans/configure`	Assessment configuration console with Target Scope selector, Engine checkboxes (SAST, DAST, KG, Docker), and Authorization checklist.	• **"Launch Assessment"** button dispatches scan. • Target dropdown bound to authorized lab (`:8001`). • Scope Guard warning banner enforcing local whitelist.
5	**Scan Timeline** `/console/scans/live`	Live multi-stage progress bar showing percentage completion, current stage badge (e.g. `VALIDATE`), and streaming console logs.	• Live progress tracking from 0% to 100%. • Real-time stage transition indicators. • One-click navigation to inspect newly discovered findings.
6	**Findings Explorer** `/console/findings`	Comprehensive vulnerability catalog with severity pills (Critical, High, Medium), CWE tags, CVSS scores, and status tags.	• Search input filtering by keyword/CWE. • Severity filter pills. • **"View Evidence"** and **"Remediate"** buttons on each finding card.
7	**Finding Detail & Evidence** `/console/findings/:id`	Forensic evidence view showing line-numbered source code with red taint sink highlights, propagation path, and CWE/OWASP mapping.	• **"Analyze with Copilot"** button. • **"Remediate in Lab"** button. • Code block with highlighted taint sink at Line 42.
8	**Security Knowledge Graph** `/console/knowledge-graph`	Interactive force-directed graph canvas connecting Application Nodes, Endpoints, Taint Sinks, CWE Definitions, and CISA KEV entries.	• Interactive node click inspection. • Zoom and pan controls. • Subgraph filter isolating specific vulnerabilities (e.g. CWE-89).
9	**AI Security Analyst** `/console/ai-analyst`	Split-screen reasoning workspace: automated exploitability breakdown on the left, interactive cybersecurity conversational terminal on the right.	• **"Inquire"** button sends prompts to AI reasoning terminal. • Displays Exploitability score, confidence level, and root cause. • **"Remediation Lab"** transition link.
10	**Remediation Workspace** `/console/remediation`	Side-by-side git diff viewer displaying vulnerable code (red) vs synthesized parameterized patch (green) with safety impact notes.	• **"Generate Patch"** calls automated synthesis. • **"Send to Sandbox"** dispatches candidate code. • Visual line-by-line diff highlighting string sanitization.

#	Page & Route	Visual Layout & Appearance	Functionalities & Interactive Buttons
11	**Sandbox Validation** `/console/sandbox`	Container validation center with 4 live check cards: Syntax, Regression Tests, AST Verification, and Security Rescan.	• **"Validate Clean Patch"** (All 4 checks turn GREEN). • **"Simulate Flawed Patch"** (Catches regression, turns RED). • **"Rollback"** button reverts flawed candidate.
12	**Posture Comparison** `/console/posture`	Quantitative before-and-after evaluation dashboard showing risk score reduction, resolved vulnerability counts, and attack surface shrinkage.	• Posture delta card demonstrating -74.2% risk drop. • Before vs After radar charts. • **"Export Evaluation JSON"** button.
13	**Security Policies** `/console/policies`	Compliance mapping matrix showing OWASP ASVS v4.0.3, CWE Top 25 (2024), and PCI-DSS v4.0 with control accordions.	• Framework tabs switching standards. • Passing controls counter (e.g. 23/30 Controls Pass). • Mapped finding links and audit traceability.
14	**Threat Intelligence** `/console/threat-intelligence`	Active threat feed stream displaying CISA Known Exploited Vulnerabilities (KEV) matches, FIRST EPSS exploit probability scores, and NVD links.	• **"Sync Feeds"** button updates live indicators. • CISA KEV alert cards for zero-day matching. • NVD advisory external links.
15	**Research Benchmark Lab** `/console/research`	Academic benchmark evaluation tables assessing detection Precision (94.2%), Recall (89.5%), F1-Score (91.8%), and FPR across 120 test cases.	• 4 Research Questions (RQ1–RQ4) status cards. • Tool comparison table (Semgrep vs ZAP vs AegisOne). • **"Export Raw Manifest"** download button.
16	**Reports & Evidence** `/console/reports`	Security report center with Executive Summary, Technical Remediation Report, and Cryptographic Evidence verification status.	• **"Download JSON"** export button. • **"Download Markdown"** export button. • SHA-256 chain verification badge.
17	**User Roles & Access** `/console/admin/access`	RBAC operator administration console with user list, assigned roles (SuperAdmin, SecurityAnalyst, Developer, Auditor), and permission matrix.	• **"+ Invite Operator"** modal dialog. • Dual-control multi-party authorization policy switch. • API key fingerprint inspection.
18	**Integration Hub** `/console/admin/integrations`	System health monitor showing status of 6 core services: FastAPI, SQLite, Semgrep, ZAP, Knowledge Graph, and Docker Engine.	• **"Probe All Services"** trigger. • Individual **"Test Connection"** buttons with live latency. • Cluster latency counter (14ms).
19	**Global Settings** `/console/admin/settings`	Defensive configuration console with Scope Guard enforcement toggles, rate limiting, and sandbox timeouts.	• Scope Guard whitelist toggle. • Session timeout controls. • **"Save Configuration"** button.
20	**Analyst Profile** `/console/account`	Operator personal workspace showing cryptographic Ed25519 signing key, assigned clearance level, and session activity timeline.	• **"Refresh Log"** updates live audit feed. • Ed25519 public key fingerprint display. • Personal triage event history.

5. Step-by-Step Live Demo Script (For Today's Review)

Follow this 5-minute presentation script during your demonstration. Speak calmly and click the buttons in this exact sequence:

STEP 1: Open the Public Landing Page (30 Seconds)

Action: Open browser at `http://localhost:5173`.

> *"Good morning, respected examiners. This is **AegisOne AI**, an explainable DevSecOps copilot. As you can see on our landing page, our core philosophy is: Security fixes should be proven safe before they ship. We don't just detect vulnerabilities; we validate that remediation patches actually work in an isolated sandbox without breaking existing features."*

Action: Click the red button **"Open Security Command Center"**.

STEP 2: Command Center & Launching a Scan (45 Seconds)

Action: Shows `http://localhost:5173/console`. Click **"Configure Scan"** on the sidebar.

> *"Here is our Security Command Center. Notice our **Scope Guard** is actively armed on our local College Placement Portal target at port 8001. Now, let's configure an assessment. AegisOne combines Static AST analysis, Dynamic active fuzzing, and Knowledge Graph corroboration."*

Action: Check the explicit authorization box and click **"Launch Assessment"**.

STEP 3: Live Scan Execution Stream (30 Seconds)

Action: Screen navigates to `/console/scans/live`. The progress bar advances through stages.

> *"The scan executes through our 10-stage pipeline: Detect, Contextualize, Prioritize, and Synthesize. Once complete, findings are normalized and mapped to CWE classifications."*

Action: Click **"Vulnerability Findings"** on the sidebar.

STEP 4: Finding Detail & AST Taint Trace (45 Seconds)

Action: Click on **"SQL Injection in Student Search Endpoint"** (CWE-89) at `/console/findings/:id`.

> *"Here is our forensic evidence. Notice that AegisOne does not just give a generic warning. It displays the exact source code taint sink at **Line 42** where unvalidated user input flows directly into dynamic SQL execution. It also cross-references this with our Security Knowledge Graph."*

Action: Click **"Analyze with Copilot"** to jump to `/console/ai-analyst`.

STEP 5: AI Security Analyst & Remediation Diff (45 Seconds)

Action: Shows `/console/ai-analyst` and then clicks **"Remediation Workspace"** (`/console/remediation`).

> *"Our AI Analyst identifies the root cause and exploitability score. In the Remediation Workspace, it synthesizes a deterministic code fix: replacing string concatenation with safe parameterized bindings. But here is where traditional tools stop and AegisOne shines: we don't trust the AI blindly."*

Action: Click **"Sandbox Validation"** on the sidebar (`/console/sandbox`).

★ STEP 6: THE SHOWSTOPPER — Sandbox Validation & Auto-Rollback (60 Seconds)

Action: On `/console/sandbox`, click **"Validate Clean Patch"**.

> *"In our Sandbox Validation Center, the candidate patch is deployed inside an isolated container. It executes 4 validation gates: Syntax Check, Unit Tests, AST Taint Check, and Security Rescan. Notice all 4 gates turn **GREEN (PASSED)**. The patch is certified 'Production-Eligible'."*

Action: Now click **"Simulate Flawed Patch"**.

> *"Now watch what happens when an AI generates a flawed patch that breaks user authentication. AegisOne runs the regression suite, detects the contract failure, turns **RED (FAILED)**, and immediately triggers an **Automated Rollback**! This guarantees unsafe AI code never breaks production."*

STEP 7: Cryptographic Proof & Conclusion (30 Seconds)

Action: Click **"Reports & Evidence"** (`/console/reports`) and point to the SHA-256 chain badge.

> *"Finally, every action is sealed into an immutable SHA-256 hash-chained audit log with Ed25519 digital signatures for zero-trust compliance. In conclusion, AegisOne AI bridges the gap between vulnerability discovery and safe, verified deployment. Thank you, and we welcome your questions!"*

6. Viva & Review Defense Questions & Answers (Memorize These!)

Q1: "Why not just use SonarQube or standard static analysis tools?"

Answer: "SonarQube is a passive scanner that only checks code syntax and produces raw alert lists without understanding real runtime exposure or data sensitivity. More importantly, SonarQube cannot generate verified patches or test them in a live sandbox. AegisOne is an active, closed-loop copilot that detects, contextualizes with a Knowledge Graph, fixes, and sandbox-validates the patch."

Q2: "Why can't developers just copy-paste code into ChatGPT to fix vulnerabilities?"

Answer: "Studies show that over 40% of AI-generated code contains subtle bugs or security regressions. ChatGPT generates code that *looks* convincing, but it has no execution environment to verify if the fix broke existing application logic. AegisOne isolates the AI patch in a disposable container and runs unit regression tests and AST verification before allowing any merge."

Q3: "What is your main academic / research contribution?"

Answer: "Our research contribution is fourfold: (1) Graph-based contextual risk scoring combining SAST and DAST, (2) Evidence-grounded AI reasoning that eliminates hallucination, (3) A 4-stage containerized validation engine with automated contract-preserving rollback, and (4) An immutable SHA-256 hash-chained audit trail providing mathematical proof of compliance."

Q4: "What is Scope Guard and why is it necessary?"

Answer: "Scope Guard is an architectural security interceptor. When an AI agent or automated scanner runs, there is a risk it could send fuzzing payloads to public third-party APIs or unintended servers. Scope Guard strictly enforces a network whitelist—blocking all outbound requests except to approved localhost and test lab containers."

Q5: "How does the Knowledge Graph improve prioritization?"

Answer: "Standard CVSS assigns a score in isolation. Our Knowledge Graph connects code endpoints to data sensitivity (e.g. Student PII) and network exposure (e.g. Public vs Internal). An SQL injection on a public endpoint accessing private data receives a 92.1 score, while an injection on a local internal script receives lower priority. This eliminates developer alert fatigue."

Q6: "How do you prove your audit trail hasn't been tampered with?"

Answer: "We use a cryptographic hash chain similar to blockchain principles. Each event's hash is calculated using the previous event's hash: `H_n = SHA256(H_{n-1} + Data_n)`. If anyone alters even a single character in past logs, the hash chain breaks immediately. Our `/api/v1/audit/verify-chain` endpoint mathematically validates the integrity of all blocks."

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